# calyr.ai.syntax

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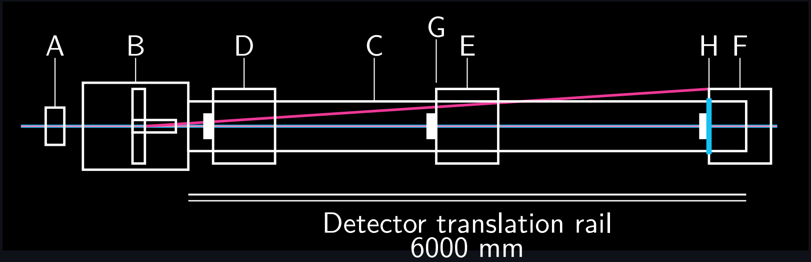
# Instrument Layout




A: Collimators. B: Sample chamber with two sample positions. C: 6 m vacuum flight tube.

D: WAXS detector (beamstop). E: SAXS detector (beamstop). F: USAXS detector (beamstop).

G: PSL (Primary Scattering Line). H: Cyan Q-vector (scattering direction marker).

Cyan = direct X-ray beam. Magenta = scattering trajectory. Detectors translate along the 6000 mm rail to set the accessible $q$-range.

# FAIR Scientific Data Problem

- Scientific datasets are usually stored at different semantic levels: raw signals, fitted curves, metadata, models, and loose analysis files.
- That makes them hard to compare, hard to reuse, and difficult to keep FAIR once they move between tools.
- The central problem is not only storage. It is preserving the scientific object across heterogeneous datasets and workflows.

**Pressure points**

Findable data.
Comparable semantics.
Reusable workflows.

# Core Idea: One Abstraction Layer

## Definition

The goal is one abstraction layer in which heterogeneous scientific datasets become typed scientific objects before they become scripts, engine inputs, or analysis fragments.

**FAIR**

Keep signal, metadata, model context, and provenance together.

**ALIGN**

Map SAXS, SPR, and other modalities to one semantic level.

**NEXUS**

Only then bind those objects to Nexus, execution, and evaluation.

# Why Abstraction Matters

## SAXS

Signal lives in reciprocal space: measured $I(q)$, optional $P(r)$, metadata, and model attachments all describe one structural sample object.

- structure-sensitive observables
- cohort-scale comparison across many entries
- representation learning on aligned curves

## SPR

Signal lives in the time and concentration domain: kinetics, equilibrium occupations, fit parameters, and conditions still describe one experimental sample object.

- dynamic process observables
- state occupation and binding interpretation
- same need for typed, reusable semantics

# SAXS as FAIR Scientific Object

## SAXS example

Public SASBDB entries such as SASDXF6, SASDYY5, and SASDC44 already combine measured $I(q)$, optional $P(r)$, metadata, and model files in one reproducible object layout.

- The first stable representation is the measured scattering curve, not an isolated downstream derivative.
- Optional model and $P(r)$ files remain attached as context, not mandatory preprocessing.
- This is already close to FAIR thinking: one identifiable sample object with reusable scientific context.

# SPR as FAIR Scientific Object

### Time traces

Measured: response vs. time
Context: concentration, temperature, surface state
Role: dynamic adsorption observable

### Equilibrium

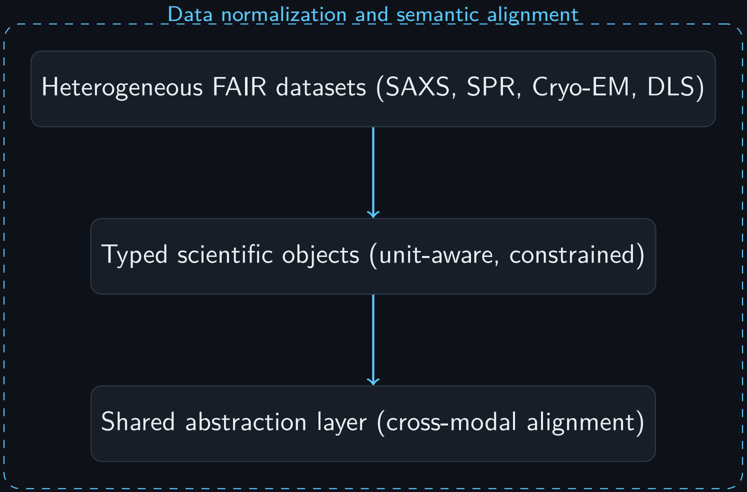
Measured: steady-state occupation
Context: concentration dependence
Role: binding and saturation interpretation

### Fit parameters

Measured: rates, occupancies, coupling terms
Context: model assumptions
Role: reusable process description

SAXS and SPR differ at the measurement surface, but both need one typed scientific object that carries signal, context, and downstream interpretation together.

# From FAIR Data to Structured Scientific Objects



# Nexus Concept

**Definition**

Nexus is the operational layer that preserves the meaning of the scientific object while it moves through execution, transformation, and evaluation.

- It binds one object graph to environments, engines, inputs, and outputs.
- It keeps FAIR data and analysis connected instead of letting them fragment into file-level side products.
- It records each transition as a manifest, so execution remains inspectable and reusable.

# PCA Intro for Both Regimes

### Input to PCA

A collection of samples is mapped to a common numerical representation, for example downloaded SAXS curves on a fixed $q$-grid or aligned feature vectors from IgG simulation and analysis.

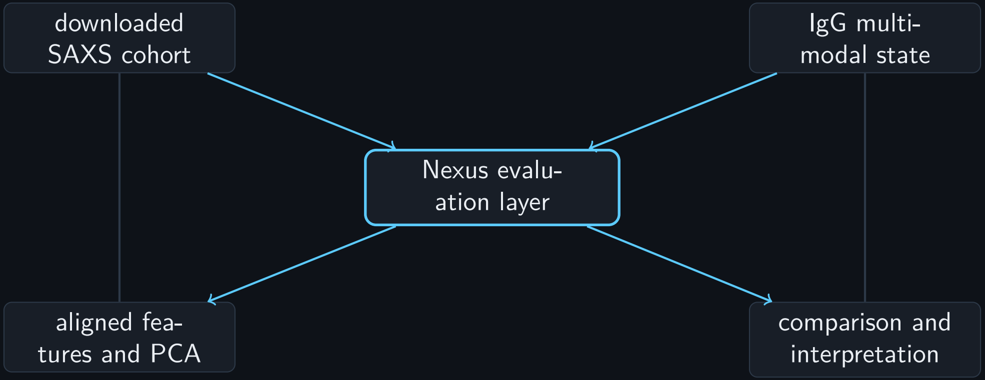
- one row per protein entry, state, frame, or condition
- one aligned feature vector per row
- PCA reveals dominant structural or observable variation modes

### Interpretation

The principal components are not new raw data. They are a compact coordinate system for comparing measured cohort samples and structured IgG states without breaking sample identity.

In the repo workflow, PCA is only meaningful once observables or fitted parameter sets are dimensionally aligned.

# Representation Flow




Nexus holds both broad downloaded cohorts and deeply structured multimodal samples, so new setups can enter the same evaluation plane without redesigning the language each time.

# One Evaluation Level

### Without Nexus

Model, script, raw output, and analysis drift into separate layers with different naming and different state.

- hard to compare runs
- analysis detached from provenance
- semantics collapse into files

### With Nexus

The same semantic object is carried from setup to result, so data evaluation stays on one coordinated plane.

- one identity for run and result
- one manifest chain for provenance
- one level for simulation and analysis

# Why We Do Not Need Dmax

Key point

Dmax belongs to the optional real-space $P(r)$ description. It is not required to place downloaded SAXS curves or IgG SAXS observables into a Nexus representation or to run PCA across aligned samples.

- PCA can operate directly on standardized $I(q)$ vectors.
- PCA can also operate on fitted parameter vectors when they have common length.
- Dmax is useful for $P(r)$ interpretation and sanity checks, but it is not a gatekeeper for representation learning.
- This keeps the evaluation layer closer to the measured data and avoids introducing an unnecessary extra assumption early in the pipeline.

# The Core Problem: Observables Span Entire Domains

**SPR: time domain $R(t)$**

$t \approx 0$ $\longrightarrow$ $t \to \infty$

$$\Longrightarrow$$

**SAXS: reciprocal space $I(q)$**

$q \approx 0$ $\longrightarrow$ $q \to \infty$

**Real space: pair distribution $p(r)$**

$r = 0$ $\longrightarrow$ $r \to \infty$

No local interpretation is sufficient
All observables depend on the full domain

Key insight

# From Model Fitting to PCA-Based Prediction

## Abstract observables

SPR is observed as a weighted response superposition:

$$R(t, C) = R_0(C) + \sum_{i=1}^{2} Q_i \theta_i(t, C)$$

SAXS is observed through the mixed pair-distance distribution and its Fourier projection:

$$PDD_{\mathrm{frac}}(r) \approx \sum_{j=1}^{M} f_j P_j(r), \qquad I(Q) = \mathrm{FFT}(PDD_{\mathrm{frac}}(r))\,[Q]$$

**Both modalities require one model spanning the full domain, not local point-wise interpretation.**

# From Model Fitting to PCA-Based Prediction

## SAXS observable from state fractions

For structural states with fractions $f_j$ and $\sum_j f_j = 1$,

$$PDD_{\mathrm{frac}}(r) \approx \sum_{j=1}^{M} f_j P_j(r), \qquad I_{\mathrm{mix}}(Q) = \mathrm{FFT}(PDD_{\mathrm{frac}}(r))\,[Q]$$

$$I_j(Q) = \mathrm{FFT}(P_j(r))\,[Q]$$

SAXS gives the Fourier-space signature of the weighted state mixture.

# From Model Fitting to PCA-Based Prediction

## Proposed extension

Use PCA across aligned features from both modalities:

$$\{I_{\mathrm{mix}}(Q)\}_{\mathsf{SAXS}}, \quad \{R(t)\}_{\mathsf{SPR}}, \quad \{\theta\text{-parameters}, f_j\}$$

**Goal: fast prediction instead of repeated fitting.**

# From Model Fitting to PCA-Based Prediction

## What PCA should learn

The representation should encode both measured dynamics and structure-derived mixtures on one aligned domain.

- Build a joint representation of protein state, kinetics, and mixture composition
- Map experimental systems into one PCA space instead of fitting each case in isolation
- Predict adsorption behavior, structural class, and multi-binding regimes

## Key idea

SAXS provides Fourier-space structural constraints from the weighted fractions.
SPR provides kinetics and equilibrium occupations.
PCA provides the predictive layer.

# Wrap-Up and Open Items

## Current status

The concept deck now links symbolic language, Nexus execution, SPR kinetics, SAXS observables, and PCA-based prediction in one narrative.

- Code and homepage presentation tested locally
- ASC transfer and operational rollout in the coming days
- Targeted improvements to the presentation and execution workflow remain open

**Next step: move from concept validation to operational transfer and refinement.**